AI HR Interview Simulator

Detailed Working Methodology, Free Tools Architecture & Version 1 Execution Plan

Complete Operational Flow | Zero Cost ($0) Tech Stack | Android Native + Groq AI | V1 Execution Plan

1. What is this App? (App Kya Hai Aur Kya Kam Karega?)

Overview: Yeh app ek **Real-time AI Executive Interview Simulator** hai. User mobile ka front camera open karke ek realistic 3D AI HR Avatar ke samne baithkar 15-30 min ka live interview de sakta hai. AI Candidate ki voice aur front camera ke zariye unke Confidence, Eye-Contact, aur Smile Score ko real-time tracking karega.

2. Complete App Working Flow (App Kaise Kaam Karega?)

1. **Step 1: Onboarding & Profile Setup:** User app mein apna Name, Target Job Role (Software, Banking, Corporate), Experience level, aur Resume Text / Certificates upload karega (Real ya Fake records support honge).
2. **Step 2: HR Character & Company Selection:** User 4-5 Avatar Characters mein se apna interviewer select karega (e.g., Rohan - Friendly Tech HR, Ananya - Corporate HR, Vikram - Strict FAANG Lead). Company select karega (Google, Microsoft, JP Morgan, ya Normal Practice).
3. **Step 3: The Virtual Desk Room:** User ka camera open hoga. Screen par Avatar office desk aur chair par baitha dikhega. AI HR table par rakhe User Profile/Certificates read karne ka gesture karega.
4. **Step 4: Real-time Live Interview & Vision Sensing:** AI HR naturally bolna start karega. Jab user jawab dega, phone ka Front Camera MediaPipe AI se candidate ka Eye Contact, Head Balance, aur Smile Score tracking karega.
5. **Step 5: Result, Certificate & Social Virality:** Interview pass hone par AI user ko overall score ke sath downloadable Verified Selection Certificate dega (jo LinkedIn/WhatsApp status par flex hoga). Reject hone par dynamic HR rejection feedback recording milegi.

3. Version 1 (MVP) $0 Free Tools Stack

Module	Selected Technology / API	Why This Tool?	Cost
AI Brain (LLM)	Groq Cloud API (Llama 3 70B)	Ultra-fast inference speed, Zero-lag replies, Multilingual (Hi/En/Mr).	100% FREE
Vision Sensing	Google MediaPipe (Android SDK)	Tracks 468 face landmarks, Eye Contact, Smile % directly on device CPU.	100% FREE
Voice Processing	Android Native STT & TTS	Built-in Android Speech-to-Text & Text-to-Speech engines for zero API cost.	100% FREE
Avatar Display	Seamless Video Loops + Canvas Overlay	Human-like desk/chair background animation with low-latency lip-sync visual.	100% FREE
AI Chat Hub	Google Gemini API (Free Tier)	1,500 free daily requests for text Q&A practice & resume tips.	100% FREE
Certificate Engine	Android Native Canvas Bitmap	Auto-generates HD dynamic image certificates directly on phone memory.	100% FREE

4. Pricing & Monetization Model

FREE TIER

₹0 Always

- 1 Mock Interview/Day
- Basic Avatars

DAY PASS

₹29 / 24 Hours

- Unlimited Interviews for 1 day
- All Companies & Avatars

PRO MONTHLY

₹199 / Month ($9.99 Global)

- Unlimited Access
- FAANG & Bank PO Modes

• Watermarked Certificate	• Full Vision Analytics	• Clean Verified Certificates

5. Version 1 Execution Plan (4-Week Development Roadmap)

- **Week 1 (UI & Profile Setup):** Android App Project setup in Jetpack Compose, User Onboarding Form, Resume/Certificate Parser.
- **Week 2 (Vision Sensor Engine):** Camera Preview integration with Google MediaPipe Face Mesh for live Confidence & Eye-Contact score.
- **Week 3 (Groq AI & Voice Pipeline):** Groq Llama-3 API integration, Android Native Voice Recognition & Text-To-Speech pipeline, Character Selection setup.
- **Week 4 (Certificates, Chat & Launch):** Dynamic Certificate Generator, Practice AI Chat Hub tab, Closed Beta testing, and Google Play Store submission.